

MIT 6.1200J Mathematics for Computer Science

Problem Set Solutions

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Part I

Problem Set 1

Problem 1: Collaboration Policy

Problem 2: Set Formulas and Propositional Formulas

(a) Verify that $(P \wedge \bar{Q}) \vee (P \wedge Q)$ is equivalent to P using a truth table.

P	Q	$\bar{Q}$	$P \wedge \bar{Q}$	$P \wedge Q$	$(P \wedge \bar{Q}) \vee (P \wedge Q)$	P
T	T	F	F	T	T	T
T	F	T	T	F	T	T
F	T	F	F	F	F	F
F	F	T	F	F	F	F

The column for $(P \wedge \bar{Q}) \vee (P \wedge Q)$ agrees with the column for P in every row, so the two formulas are equivalent.

(b) Prove that $A = (A - B) \cup (A \cap B)$ for all sets A, B , by showing $x \in A$ iff $x \in (A - B) \cup (A \cap B)$ for all elements x .

Proof. Let x be arbitrary.

($\Rightarrow$) Suppose $x \in A$. Either $x \in B$ or $x \notin B$.

- If $x \notin B$, then $x \in A$ and $x \notin B$, so $x \in A - B$, hence $x \in (A - B) \cup (A \cap B)$.
- If $x \in B$, then $x \in A$ and $x \in B$, so $x \in A \cap B$, hence $x \in (A - B) \cup (A \cap B)$.

Either way, $x \in (A - B) \cup (A \cap B)$.

($\Leftarrow$) Suppose $x \in (A - B) \cup (A \cap B)$. Then $x \in A - B$ or $x \in A \cap B$.

- If $x \in A - B$, then by definition $x \in A$ (and $x \notin B$).
- If $x \in A \cap B$, then $x \in A$ (and $x \in B$).

Either way, $x \in A$.

Since $x \in A$ iff $x \in (A - B) \cup (A \cap B)$ for arbitrary x , the two sets are equal: $A = (A - B) \cup (A \cap B)$. $\square$

Relation to part (a). This is the set-theoretic mirror of part (a). Substitute $P := "x \in A"$ and $Q := "x \in B"$. Then:

- $x \in A - B$ corresponds to $P \wedge \bar{Q}$ (in A , not in B),
- $x \in A \cap B$ corresponds to $P \wedge Q$ (in A , in B),
- $\cup$ corresponds to $\vee$.

So the set identity $A = (A - B) \cup (A \cap B)$ is exactly the membership-level statement $P \leftrightarrow ((P \wedge \bar{Q}) \vee (P \wedge Q))$ from part (a), and the case analysis in the proof above is just an unfolding of the truth table: the two cases $x \notin B$ / $x \in B$ are the two ways Q can be false or true, matching the two rows of the truth table where P is true.

Problem 3: Predicate Practice

- (a) The predicate $\text{sixish}(n)$, “ n is a natural number whose first digit is 6”:

$$\text{sixish}(n) := \exists k \in \mathbb{N}. (6 \cdot 10^k \leq n < 7 \cdot 10^k).$$

- (b) “There are exactly two natural numbers n such that $Q(n)$ is true”:

$$\exists m, n \in \mathbb{N}. (m \neq n \wedge \forall k \in \mathbb{N}. (Q(k) \leftrightarrow (k = m \vee k = n))).$$

- (c) Goldbach’s Conjecture: “every even integer greater than 2 can be written as the sum of two primes.” Following the required form $\forall n. W(n)$ with all the logic folded into W :

$$\forall k \in \mathbb{N}. \left[(k > 2 \wedge \text{isEven}(k)) \rightarrow \exists m, n \in \mathbb{N}. (\text{isPrime}(m) \wedge \text{isPrime}(n) \wedge k = m + n) \right],$$

where, since isEven was not predefined for us, we take $\text{isEven}(k) := \neg \text{isOdd}(k)$.

- (d) “Every prime larger than 100 is sixish”:

$$\forall n \in \mathbb{N}. \left[(n > 100 \wedge \text{isPrime}(n)) \rightarrow \text{sixish}(n) \right].$$

- (e) Is $R(n) := \exists b \in \mathbb{N}. (n = 2b + 5)$ equivalent to $\text{isOdd}(n)$?

Claim: No, R is not equivalent to isOdd .

Proof. Suppose, toward a contradiction, that $R(n) \leftrightarrow \text{isOdd}(n)$ for all $n \in \mathbb{N}$. Take $n = 3$. Since 3 is odd, $\text{isOdd}(3)$ is true, so by our assumption $R(3)$ must also be true. But

$$R(3) := \exists b \in \mathbb{N}. (3 = 2b + 5),$$

and solving $3 = 2b + 5$ gives $b = -1 \notin \mathbb{N}$, so no such b exists and $R(3)$ is false. This contradicts $R(3)$ being true. Hence our assumption was wrong, and R is not equivalent to isOdd .

(Intuitively, $R(n)$ only captures odd numbers $n \geq 5$, since b ranges over $\mathbb{N}$; it misses the smaller odd numbers 1 and 3.) □

Problem 4: Induction and Contradiction

- (a) *Setup.* $Q(x)$ is a predicate on $\mathbb{R}$ such that for all $x, y \in \mathbb{R}$, if $Q(x \cdot y)$ is true then $Q(x)$ or $Q(y)$ is true. We show the stronger property: for every integer $n \geq 1$ and all $x_1, \dots, x_n \in \mathbb{R}$, if $Q(x_1 x_2 \cdots x_n)$ is true then at least one of $Q(x_1), \dots, Q(x_n)$ is true.

Proof. Let $P(n)$ be the statement: for all $x_1, \dots, x_n \in \mathbb{R}$,

$$Q(x_1 x_2 \cdots x_n) \rightarrow Q(x_1) \vee Q(x_2) \vee \cdots \vee Q(x_n).$$

We prove $P(n)$ holds for all integers $n \geq 1$ by induction on n .

Base case ($n = 1$): $P(1)$ says: for all $x_1 \in \mathbb{R}$, $Q(x_1) \rightarrow Q(x_1)$, which is trivially true.

Inductive step: Suppose $n \geq 1$ and $P(n)$ holds; we show $P(n+1)$ holds. Let $x_1, \dots, x_n, x_{n+1} \in \mathbb{R}$ be arbitrary, and suppose $Q(x_1 x_2 \cdots x_n x_{n+1})$ is true. Write

$$y := x_1 x_2 \cdots x_n, \quad z := x_{n+1},$$

so that $x_1 \cdots x_n x_{n+1} = yz$. By hypothesis, $Q(yz)$ is true. Applying the given property of Q (i.e., $P(1)$ is not what we need here — we use the defining property of Q directly, which is exactly $P(2)$ applied to y, z) to y and z :

$$Q(yz) \rightarrow Q(y) \vee Q(z).$$

Since $Q(yz)$ holds, we conclude $Q(y) \vee Q(z)$ holds, i.e.,

$$Q(x_1 x_2 \cdots x_n) \vee Q(x_{n+1}).$$

Now consider the two cases:

- If $Q(x_{n+1})$ holds, then in particular $Q(x_1) \vee \cdots \vee Q(x_n) \vee Q(x_{n+1})$ holds.
- If $Q(x_1 x_2 \cdots x_n)$ holds, then by the inductive hypothesis $P(n)$ applied to $x_1, \dots, x_n$, we get

$$Q(x_1 x_2 \cdots x_n) \rightarrow Q(x_1) \vee \cdots \vee Q(x_n),$$

so $Q(x_1) \vee \cdots \vee Q(x_n)$ holds, and hence so does $Q(x_1) \vee \cdots \vee Q(x_n) \vee Q(x_{n+1})$.

In either case, $Q(x_1) \vee \cdots \vee Q(x_n) \vee Q(x_{n+1})$ holds, which is exactly $P(n+1)$.

By mathematical induction, $P(n)$ holds for all integers $n \geq 1$. □

Remark (fixing a gap). The original argument substituted y and z back into $Q(y) \vee Q(z)$ and asserted the result followed “by the inductive hypothesis,” but never explicitly invoked $P(n)$ on $Q(y) = Q(x_1 \cdots x_n)$. The step above makes that invocation explicit: we need $P(n)$ to expand $Q(x_1 \cdots x_n)$ into $Q(x_1) \vee \cdots \vee Q(x_n)$.

(b) Theorem 1. If the product of $n \geq 1$ real numbers is irrational, then at least one of those numbers must be irrational.

To derive this from part (a), define

$$Q(x) := “x \text{ is irrational}.”$$

Before part (a) applies to this Q , we must check Q satisfies the hypothesis of part (a), namely the base property

$$\forall x, y \in \mathbb{R}. (Q(xy) \rightarrow Q(x) \vee Q(y)).$$

Equivalently (by contrapositive), we must prove the following lemma:

Lemma. For all $x, y \in \mathbb{R}$, if x and y are both rational, then xy is rational.

Once the Lemma is established, part (a) (with this Q) gives exactly Theorem 1.

(c) Proof of the Lemma, by contradiction.

Proof. Suppose, toward a contradiction, that x and y are both rational but xy is irrational. Since x and y are rational, by definition there exist $p_1, q_1, p_2, q_2 \in \mathbb{Z}$ with $q_1 \neq 0, q_2 \neq 0$, such that

$$x = \frac{p_1}{q_1}, \quad y = \frac{p_2}{q_2}.$$

Then

$$xy = \frac{p_1}{q_1} \cdot \frac{p_2}{q_2} = \frac{p_1 p_2}{q_1 q_2}.$$

Since $p_1, p_2, q_1, q_2 \in \mathbb{Z}$, we have $p_1 p_2 \in \mathbb{Z}$ and $q_1 q_2 \in \mathbb{Z}$; moreover $q_1 q_2 \neq 0$ since $q_1 \neq 0$ and $q_2 \neq 0$. Hence xy is expressible as a ratio of two integers with nonzero denominator, i.e. xy is rational. This contradicts our assumption that xy is irrational. Therefore, no such x, y exist: whenever x and y are both rational, xy is rational. □

Finishing the proof of Theorem 1. The Lemma is precisely the contrapositive of the required property of $Q(x) := “x \text{ is irrational},”$ namely

$$\forall x, y \in \mathbb{R}. Q(xy) \rightarrow Q(x) \vee Q(y).$$

(Indeed: if xy is irrational, then x, y cannot both be rational, by the Lemma; so at least one of x, y is irrational.) Since Q satisfies the hypothesis of part (a), part (a) tells us that for every integer $n \geq 1$ and all $x_1, \dots, x_n \in \mathbb{R}$,

$$Q(x_1 x_2 \cdots x_n) \rightarrow Q(x_1) \vee \cdots \vee Q(x_n),$$

i.e., if $x_1 x_2 \cdots x_n$ is irrational, then at least one of $x_1, \dots, x_n$ is irrational. This is exactly Theorem 1. ■

Part II

Problem Set 2

Problem 1: Outlining Proofs

(a) Theorem: “Every natural number is tall or wide, but not both.” (Proof method: proof by cases.)

Writing $T(n)$ for “ n is tall” and $W(n)$ for “ n is wide,” the theorem says

$$\forall n \in \mathbb{N}. (T(n) \oplus W(n)),$$

where $\oplus$ is exclusive or. We outline a proof by cases on whether n is tall or wide (this case split itself needs $T(n) \vee W(n)$ to be already known, e.g. from an earlier result — we flag this as a TODO as well).

Proof Outline. Let $n \in \mathbb{N}$ be arbitrary. [TODO: show $T(n) \vee W(n)$ (needed so the two cases below are exhaustive)]. We consider two cases.

Case 1: $T(n)$ holds. We must show $T(n) \oplus W(n)$, i.e. (since $T(n)$ is true) that $W(n)$ is false. [TODO: show $T(n) \rightarrow \neg W(n)$].

Case 2: $W(n)$ holds. Symmetrically, we must show $T(n)$ is false. [TODO: show $W(n) \rightarrow \neg T(n)$].

In both cases $T(n) \oplus W(n)$ holds, completing the proof by cases.

(b) Theorem: “There exists a happy natural number between 10 and 1000 that has no happy proper divisors.” (Proof method: choose 25 as the example.)

Writing $H(n)$ for “ n is happy” and $D(n)$ for the set of proper divisors of n , the theorem says

$$\exists n \in \mathbb{N}. \left(10 \leq n \leq 1000 \wedge H(n) \wedge \forall d. (d \mid n \wedge d < n \rightarrow \neg H(d)) \right).$$

Proof Outline. Take $n = 25$.

$10 \leq 25 \leq 1000$: true by inspection.

[TODO: show $H(25)$, i.e. that 25 is happy].

The proper divisors of 25 are 1 and 5. [TODO: show $\neg H(1)$ and $\neg H(5)$].

Since 25 is in range, happy, and has no happy proper divisor, $n = 25$ witnesses the existential claim.

(c) Theorem: “There are no devious natural numbers.” (Proof method: regular induction, with base cases 0 and 1.)

Writing $D(n)$ for “ n is devious,” the theorem says $\forall n \in \mathbb{N}. \neg D(n)$. Let

$$P(n) := \neg D(n).$$

Two base cases are called for, which suggests the (unstated) recursive structure of “devious” relates n to its two predecessors $n - 1, n - 2$ (as in Problem 2). We outline accordingly:

Proof Outline. We prove $P(n)$ for all $n \in \mathbb{N}$ by induction on n , with two base cases and a two-term inductive step.

Base case $n = 0$: *[TODO: show $P(0)$, i.e. $\neg D(0)$].*

Base case $n = 1$: *[TODO: show $P(1)$, i.e. $\neg D(1)$].*

Inductive step: Let $k \geq 1$, and assume $P(k-1)$ and $P(k)$ (i.e. $\neg D(k-1)$ and $\neg D(k)$) both hold. *[TODO: show $P(k+1)$, i.e. $\neg D(k+1)$, using the definition of devious and the two inductive hypotheses].*

By induction (using both base cases as anchors), $P(n)$ holds for all $n \in \mathbb{N}$.

Problem 2: Spotting Errors

Recall $x_1 = 4$, $x_2 = 13$, and $x_i = 6x_{i-1} - 8x_{i-2}$ for $i \geq 3$, so $(x_1, x_2, x_3, x_4, \dots) = (4, 13, 46, 172, \dots)$. Goal: $x_i > 0$ for all $i \geq 1$.

(a) Befuddled Brynmor’s mistake.

Brynmor uses strong induction on $Q(i) := “x_i \geq 2^i”$, and in the inductive step writes

$$x_{i+1} = 6x_i - 8x_{i-1} \geq 6 \cdot 2^i - 8 \cdot 2^{i-1},$$

substituting the lower bounds $x_i \geq 2^i$ and $x_{i-1} \geq 2^{i-1}$ directly into the expression.

This step is invalid. The term $6x_i$ has a positive coefficient, so substituting the lower bound $x_i \geq 2^i$ correctly gives $6x_i \geq 6 \cdot 2^i$. But the term $-8x_{i-1}$ has a *negative* coefficient: to lower-bound $-8x_{i-1}$ we need an *upper* bound on x_{i-1} , not a lower bound. Knowing $x_{i-1} \geq 2^{i-1}$ only tells us $-8x_{i-1} \leq -8 \cdot 2^{i-1}$ — the inequality goes the wrong way to conclude anything about x_{i+1} being large. So the displayed inequality $x_{i+1} \geq 6 \cdot 2^i - 8 \cdot 2^{i-1}$ does not actually follow from the inductive hypothesis, and the rest of the argument is unjustified.

(b) Erroneous Erik’s mistake.

Erik uses (regular) induction on $R(i) := “x_i \geq 3x_{i-1}”$ for $i \geq 2$. In the inductive step, after correctly deriving $x_{i+1} \geq \frac{10}{3}x_i$ from the inductive hypothesis $R(i)$, he writes: “Since $x_i > 0$, $\frac{10}{3}x_i > 3x_i$ ” in order to conclude $x_{i+1} \geq 3x_i$, i.e. $R(i+1)$.

This step is circular. The inductive step for $R(i+1)$ is only allowed to use the inductive hypothesis $R(i)$ (namely $x_i \geq 3x_{i-1}$) — it does *not* give us $x_i > 0$ for free. Positivity of x_i is exactly the fact the whole exercise is trying to establish, and it is only derived *afterward*, by combining $R(i)$ with $x_1 > 0$. Using “ $x_i > 0$ ” inside the proof of $R(i) \Rightarrow R(i+1)$, before positivity has been established, assumes the very conclusion the argument is building toward.

(c) Overzealous Zach’s mistake.

Zach uses strong induction on $S(i) := “x_i = 4^i”$, with a single base case $S(1)$, and an inductive step that (for general $i \geq 1$) invokes both $S(i-1)$ and $S(i)$ to derive $S(i+1)$ via the recurrence $x_{i+1} = 6x_i - 8x_{i-1}$.

This step fails already at $i = 1$. To derive $S(2)$ this way, the inductive step needs $S(0)$, i.e. a value $x_0 = 4^0 = 1$ — but x_0 is not defined at all (x_1, x_2 are the given base values, and the recurrence $x_i = 6x_{i-1} - 8x_{i-2}$ is only stipulated for $i \geq 3$). So the strong induction step’s use of the recurrence to compute x_2 from x_1, x_0 is illegitimate on two counts: x_0 doesn’t exist, and even x_2 itself is a *given* base value, not something produced by the recurrence. Consistent with this, the claim $S(2)$ is simply false: $x_2 = 13 \neq 16 = 4^2$. So the strong induction breaks down exactly at the step from $i = 1$ to $i = 2$.

(d) Correct proof that $x_i > 0$ for all $i \geq 1$.

Following the hint, we repair Erik’s argument by folding positivity directly into the inductive hypothesis, so it no longer needs to be smuggled in mid-proof. Let

$$P(i) := “x_i > 2x_{i-1}”, \quad \text{for } i \geq 2.$$

Proof. We prove $P(i)$ holds for all $i \geq 2$ by induction on i .

Base case ($i = 2$): $x_2 = 13$ and $2x_1 = 8$, and indeed $13 > 8$, so $P(2)$ holds.

Inductive step: Let $i \geq 2$ and assume $P(i)$, i.e. $x_i > 2x_{i-1}$. We show $P(i+1)$, i.e. $x_{i+1} > 2x_i$.

From $x_i > 2x_{i-1}$ we get $x_{i-1} < \frac{1}{2}x_i$. Multiplying both sides by -8 (which flips the inequality, since $-8 < 0$):

$$-8x_{i-1} > -8 \cdot \frac{1}{2}x_i = -4x_i.$$

Now use the recurrence $x_{i+1} = 6x_i - 8x_{i-1}$:

$$x_{i+1} = 6x_i - 8x_{i-1} > 6x_i - 4x_i = 2x_i.$$

So $x_{i+1} > 2x_i$, which is $P(i+1)$.

By induction, $P(i)$ holds for all $i \geq 2$, i.e.

$$x_i > 2x_{i-1} \quad \text{for all } i \geq 2. \quad (*)$$

Finally, we conclude $x_i > 0$ for all $i \geq 1$ by a second, simple induction, using $(*)$:

- $x_1 = 4 > 0$.
- If $x_{i-1} > 0$ for some $i \geq 2$, then by $(*)$, $x_i > 2x_{i-1} > 2 \cdot 0 = 0$.

Hence $x_i > 0$ for every $i \geq 1$, as required. □

Problem 3: Integer Multiplication from Bit Shifts

State machine on triples $(r, s, a) \in \mathbb{N}^3$, start state $(x, y, 0)$, transitions (defined only when $s > 0$):

$$(r, s, a) \mapsto \begin{cases} (2r, s/2, a) & \text{if } s \text{ is even (and } s > 0) \\ (2r, (s-1)/2, a+r) & \text{if } s \text{ is odd} \end{cases}$$

(a) Tracing from $(5, 22, 0)$ (target: $5 \cdot 22 = 110$):

$$(5, 22, 0) \rightarrow (10, 11, 0) \rightarrow (20, 5, 10) \rightarrow (40, 2, 30) \rightarrow (80, 1, 30) \rightarrow (160, 0, 110).$$

No transition applies once $s = 0$, so this is the final state, with $a_f = 110 = 5 \cdot 22$. It correctly computes the product.

(b) **Invariant.** Let $P(r, s, a) := “rs + a = xy.”$ We show P is an invariant of the state machine: true at the start state, and preserved by every transition.

Proof. Holds initially: $P(x, y, 0)$ says $xy + 0 = xy$, which is true.

Preserved by transitions: Suppose $P(r, s, a)$ holds, i.e. $rs + a = xy$, and $s > 0$ so a transition applies.

Case s even: $(r, s, a) \mapsto (2r, s/2, a)$. Then

$$(2r) \left(\frac{s}{2} \right) + a = rs + a = xy,$$

so $P(2r, s/2, a)$ holds.

Case s odd: $(r, s, a) \mapsto (2r, (s-1)/2, a+r)$. Then

$$(2r) \left(\frac{s-1}{2} \right) + (a+r) = r(s-1) + a + r = rs - r + a + r = rs + a = xy,$$

so $P(2r, (s-1)/2, a+r)$ holds.

In both cases, the successor state also satisfies P . Since P holds at the start state and is preserved by every transition, P is an invariant. $\square$

(c) **Final states.** Transitions are defined exactly when $s > 0$, so a state has no outgoing transition (is final) precisely when $s = 0$. Thus the final states are exactly the triples of the form $(r, 0, a)$ for $r, a \in \mathbb{N}$.

(d) **Partial correctness.** Let (r_f, s_f, a_f) be any reachable final state. By part (c), $s_f = 0$. Since P is an invariant (part (b)) and holds at the start state, it holds at every reachable state, including this one:

$$P(r_f, 0, a_f) : \quad r_f \cdot 0 + a_f = xy \quad \implies \quad a_f = xy.$$

So in any reachable final state, $a_f = xy$, as required.

(e) **Termination.** Let $\text{bitcount}(n) := \lceil \log_2(n+1) \rceil$ for $n \in \mathbb{N}$. We show $\text{bitcount}(s)$ strictly decreases — in fact, decreases by exactly 1 — with every transition.

It is a standard fact that for $n \geq 1$, $\text{bitcount}(n)$ is the unique integer k such that $2^{k-1} \leq n < 2^k$ (i.e. the number of bits in n 's binary representation), and $\text{bitcount}(0) = 0$.

Proof. Let (r, s, a) be any state with $s > 0$, and suppose $\text{bitcount}(s) = k$, so $2^{k-1} \leq s < 2^k$.

Case s odd: the transition sends $s \mapsto \frac{s-1}{2}$. Write $m = \frac{s-1}{2}$, so $s = 2m + 1$. From $2^{k-1} \leq s < 2^k$ we get $2^{k-1} \leq 2m + 1 < 2^k$, i.e. $2^{k-1} - 1 \leq 2m < 2^k - 1$; since $2m$ is even and $2^{k-1} - 1$ is odd (for $k \geq 1$), this sharpens to $2^{k-1} \leq 2m \leq 2^k - 2$, i.e. $2^{k-2} \leq m \leq 2^{k-1} - 1 < 2^{k-1}$. If $k \geq 2$ this says $2^{(k-1)-1} \leq m < 2^{k-1}$, so $\text{bitcount}(m) = k - 1$. (If $k = 1$, then $s = 1$, $m = 0$, and $\text{bitcount}(0) = 0 = k - 1$ by definition.) Either way,

$$\text{bitcount}\left(\frac{s-1}{2}\right) = k - 1 = \text{bitcount}(s) - 1.$$

Case s even: the transition sends $s \mapsto \frac{s}{2}$. Write $m = \frac{s}{2}$, so $s = 2m$, and (since $s > 0$) $m \geq 1$. From $2^{k-1} \leq s < 2^k$ we get $2^{k-1} \leq 2m < 2^k$, i.e. $2^{k-2} \leq m < 2^{k-1}$, so $\text{bitcount}(m) = k - 1$. Hence

$$\text{bitcount}\left(\frac{s}{2}\right) = k - 1 = \text{bitcount}(s) - 1.$$

In both cases, the new value of s has bitcount exactly one less than before. In particular, $\text{bitcount}(s)$ is a strictly decreasing, non-negative-integer-valued derived variable along any run of the machine. Since it cannot decrease forever (it is a non-negative integer), the machine cannot run forever: it must reach a state with $s = 0$ (no further transition possible), i.e. the algorithm always terminates. $\square$

(f) Time bound. By part (e), $\text{bitcount}(s)$ decreases by exactly 1 at every step, starting from $\text{bitcount}(y)$ at the initial state $(x, y, 0)$ and ending at $\text{bitcount}(0) = 0$ at the final state. Hence, the number of transitions taken is exactly

$$\text{bitcount}(y) = \lceil \log_2(y + 1) \rceil,$$

so from any starting state $(x, y, 0)$, the algorithm terminates after at most $\lceil \log_2(y + 1) \rceil$ steps.

Problem 4: Invariant Enlightenment